



with MySQL, one of the most popular relational database management systems.

Installing MySQL:

1. Download MySQL:

- Go to the MySQL website (<https://www.mysql.com/>).
- Download the MySQL Community Server for your operating system.

2. Install MySQL:

- Run the downloaded installer and follow the instructions.
- During installation, you will be prompted to configure the MySQL server, including setting a root password and choosing the server type.

3. Start MySQL Server:

- After installation, start the MySQL server from the MySQL Workbench or using the command line.

Setting Up a Database:

1. Open MySQL Workbench:

- MySQL Workbench is a visual tool for database architects and developers.
- Open MySQL Workbench and connect to your local MySQL server.

2. Create a New Database:

- In MySQL Workbench, click on the “Create a new schema” icon.
- Enter a name for your database and click “Apply”.

3. Create a Table:

- Right-click on the newly created database and select “Create Table”.
- Define the table columns, their data types, and any constraints (e.g., primary key).

Basic SQL Syntax

Understanding the basic syntax of SQL is essential for managing and manipulating data within a database. Here are some fundamental concepts and constructs:

Creating a Table:

The CREATE TABLE statement is used to create a new table in the database.

```
CREATE TABLE employees (
    id INT AUTO_INCREMENT PRIMARY KEY,
    first_name VARCHAR(50),
    last_name VARCHAR(50),
    email VARCHAR(100),
    hire_date DATE
);
```

Inserting Data:

The INSERT INTO statement is used to insert new records into a table.

```
INSERT INTO employees (first_name, last_name, email, hire_date)
VALUES ('John', 'Doe', 'john.doe@example.com', '2023-06-01');
```

Querying Data:

The SELECT statement is used to query data from a table.

```
SELECT * FROM employees;
```

```
SELECT first_name, last_name FROM employees
WHERE hire_date > '2023-01-01';
```

Updating Data:

The UPDATE statement is used to modify existing records in a table.

```
UPDATE employees
SET email = 'john.newemail@example.com'
WHERE id = 1;
```

Deleting Data:

The DELETE statement is used to remove records from a table.

```
DELETE FROM employees WHERE id = 1;
```

Altering a Table:

The ALTER TABLE statement is used to modify the structure of an existing table.

```
ALTER TABLE employees ADD COLUMN phone_number VARCHAR(20);
```

```
ALTER TABLE employees DROP COLUMN phone_number;
```

Writing and Executing SQL Queries

Let's write some SQL queries to manage data in a sample data-